PROJECT EXPLAINER

TwinRecon

From one object video to a 3D model

A beginner-friendly guide to the theory, design choices, problems solved, and current research limits.

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frames + masks

02
ALIGN
camera views

03
REBUILD
dense points

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REPRESENT
splat + mesh

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QUESTION
confidence atlas

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new view

The central question

Which parts of a good-looking reconstruction can we actually trust?

Prepared from the current TwinRecon workspace and its existing 'real' scan report.

SECTION 01

The whole project in one picture

The input is a video of one object. The output is a 3D reconstruction plus evidence about which regions are uncertain.



The key idea

TwinRecon does more than make a model. It asks whether the model is supported by the photos, masks, and a second 3D representation. A convincing render is not, by itself, proof that the shape is correct.

What each output means

Point cloud

Colored 3D samples of the object's visible surface. Points are not connected.

Gaussian splat

Many small translucent 3D ellipsoids blended into a viewable image.

Poisson mesh

Triangles form a continuous surface from the cloud's points and estimated normals.

Confidence atlas

Local scores and reason labels show where the scan needs caution or more views.

Two assumptions to say aloud

- The object should behave as one rigid object across frames. Moving or deforming parts confuse multi-view geometry.
- The video gives relative shape but no physical ruler. Without a known scale, model coordinates are arbitrary scene units.

A small vocabulary to remember

Pixel: one colored image location. **Depth:** distance from a camera to the visible surface. **Camera pose:** where a camera is and how it points. **Mask:** which pixels are the object. **Normal:** the direction perpendicular to a surface. **Floater:** a stray reconstructed fragment separated from the intended object.

SECTION 02

From video to camera geometry

The software must first infer where every image was taken. This is what makes later depth estimates possible.

Stage 1 - sample frames and isolate the object

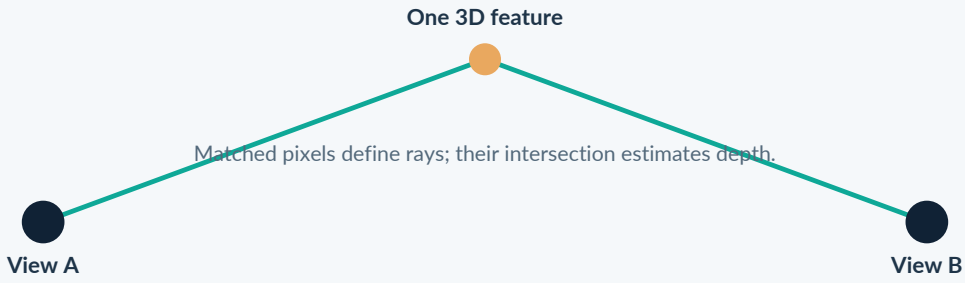
TwinRecon samples roughly 50-60 frames instead of processing every nearly identical video frame. Image size is capped to control memory. GrabCut produces an object mask for each frame, with an optional model-based segmentation route for difficult backgrounds. Masks steer reconstruction toward the object rather than the table, wall, or turntable.

Why masks matter

A background can have many more easy-to-match corners than a dark object. Without masking, a camera solver may follow the background and reconstruct the wrong scene.

Stage 2 - Structure from Motion (SfM)

COLMAP detects recognizable local features, matches the same features across frames, and estimates camera poses and a sparse cloud of 3D points. A pose is a camera's location and orientation. Bundle adjustment then changes camera poses and 3D points together to reduce their projection errors. Accurate poses are the foundation of the dense reconstruction. [1]



Triangulation in plain language: a feature seen from two known camera locations lies along two viewing rays. Their intersection estimates its 3D location. With many observations, the estimate becomes more stable.

Why dark objects are hard: low-texture and reflective regions supply fewer stable features. TwinRecon requests more SIFT features and uses a lower feature-detection threshold, but extra matches can also include mistakes. Later consistency checks are therefore essential.

The projection model

A 3D point is rotated and shifted into a camera's coordinate frame, then projected through its lens model to a 2D pixel. SfM searches for poses and points that make predicted pixels match observed feature pixels. Camera calibration accounts for focal length, image center, and lens distortion.

SECTION 03

From camera views to a clean point cloud

Sparse feature points are only a scaffold. Multi-view stereo fills in the visible surface and filtering removes likely background.

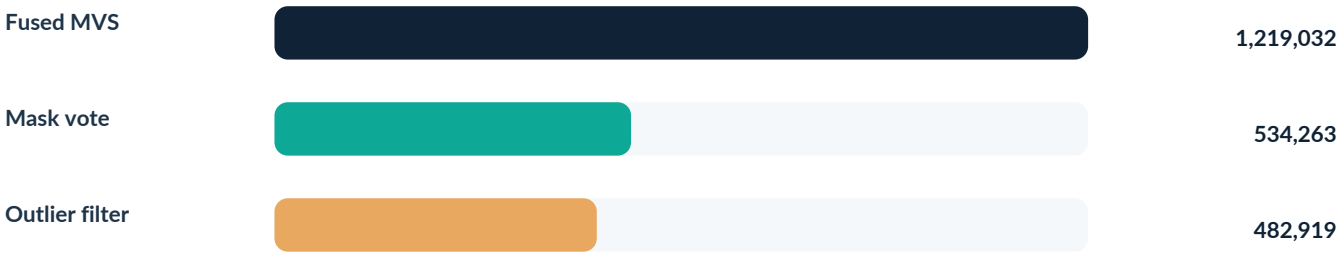
Stage 3 - dense multi-view stereo

COLMAP's PatchMatch Stereo estimates depth for image pixels by checking whether small image patches agree across registered views. Depth is the distance from a camera to a surface. Depth maps from different frames are fused into one dense point cloud. The result can contain hundreds of thousands of points. [1]

Depth fails most often on uniform black material, glare, motion blur, weak overlap, occlusions, and inaccurate masks. A dense output is not automatically a correct output.

Stage 4 - keep only well-supported object points

Each fused point is projected back into the registered frames. It survives when it is foreground in at least two views and at least 70% of its visible views. Statistical filtering removes isolated outliers. A detected plane is removed only if it is much broader than the object, which helps retain a flat shoe sole.



Worked example - the existing 'real' scan: 1,219,032 raw fused points became 534,263 after the mask vote and 482,919 after outlier filtering. The plane detector did not remove a plane in this run.

Why the conservative plane rule exists

A table and a shoe sole are both flat. A simple 'delete every large plane' rule may delete real object geometry. TwinRecon checks relative plane extent and avoids a final percentile bounding-box crop.

Two different filters: statistical outlier removal asks whether a point is isolated from neighboring points. RANSAC repeatedly tests a simple plane hypothesis to find a broad flat region. Neither can decide by itself whether a flat patch belongs to the table or the object; the mask and relative extent provide context.

SECTION 04

Three ways to represent one object

The point cloud, splat, and mesh answer different questions. None is a complete substitute for the others.

Point cloud = sampled geometry

Best for seeing where 3D measurements exist. Sparse or noisy areas remain visible. Screen dot size affects appearance but not the data.

Mesh = connected geometry

Poisson reconstruction fits a surface through oriented points. Useful for shape checks, but it may smooth details or fill unobserved holes. [3]

Gaussian splat = rendered appearance

Each point becomes a small oriented ellipsoid with color, scale, rotation, and opacity. Ellipsoids are projected to screen-space footprints and blended by depth. Local neighbor geometry makes them thin through the surface and wider along it, reducing a blobby look. [2]

Base color, spherical harmonics, and honest claims

The initial splat uses measured multi-view stereo RGB. Spherical harmonics (SH) can encode how apparent color changes with view direction. The exported PLY supports level-3 SH fields, but the existing 'real' scan has **unestimated higher-order coefficients**. Fields in a file are not proof of learned reflections. The local website currently displays base (DC) color, not view-dependent SH.

How rendering works: a 3D ellipsoid projects to an ellipse in a camera view. Its color and opacity contribute to each covered screen pixel. The renderer blends many splats from front to back; a smoother footprint can hide gaps, so visual sharpness and geometric correctness must be judged separately.

Optional short refinement

A bounded gsplat pass can adjust colors and limited shape parameters on a subset of splats at low resolution. It compares masked renders to photos, while held-out frames determine whether changes are accepted. Higher-order SH is attempted only after simpler color and shape changes improve validation views. Limits on splat count, image size, and steps protect the 4 GB GPU. [4]

Status of the current example

This refinement was added after the existing 'real' scan. The GPU driver was unavailable when that scan was checked, so no measured refinement gain or learned view-dependent reflection should be claimed for it.

SECTION 05

The confidence atlas: the research core

A region can be plausible yet poorly supported. TwinRecon grades evidence locally and explains why it distrusts an area.

The atlas groups sampled cloud points into small 3D regions. It combines four signals, each normalized to a score from 0 to 1. Their average becomes regional confidence:

Current scoring rule

Confidence = 0.25 x camera support + 0.25 x silhouette agreement + 0.25 x image agreement + 0.25 x cloud-mesh agreement. Regions below 0.8 are marked weak. These weights and threshold are design choices, not calibrated probabilities.

1 Camera support

A CPU visibility check counts useful registered views of a region. Few independent views mean less geometric certainty.

2 Silhouette agreement

Points should project inside the object's masks in the frames where they are visible.

3 Image agreement

Projected point colors are compared with video pixels. This uses a fast point-render proxy, not a full Gaussian render.

4 Cloud-mesh distance

Points far from the Poisson surface may indicate noise, a floater, or a detail smoothed away by meshing.

TwinRecon labels a weak region with its strongest reason, such as 'few camera views' or 'cloud / mesh gap'. It also flags likely floater regions and audits blurry or inconsistent frames.

Scientific caution

The mesh is built from the cloud, so their errors can be correlated. Agreement between them is internal consistency, not proof of truth. Completely missing surfaces may be absent from both and therefore have no points to score.

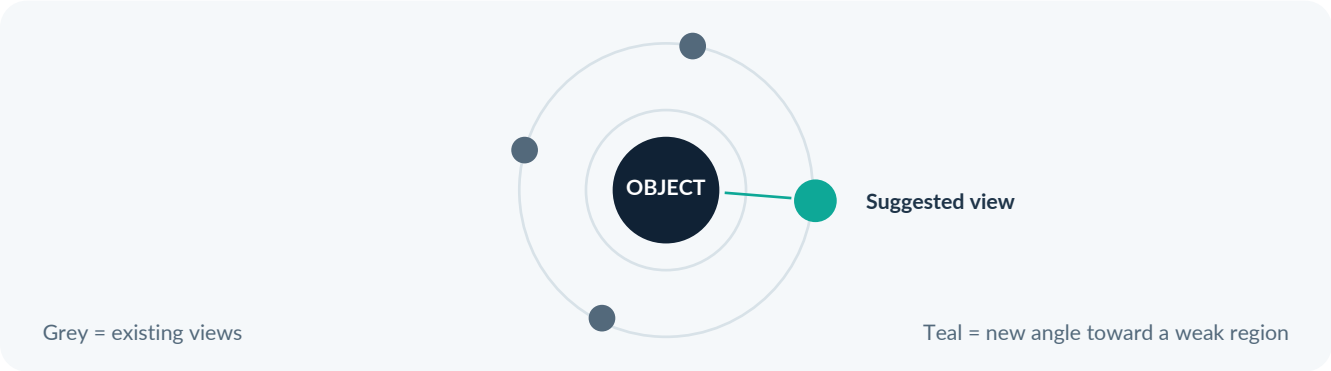
Example: if the underside was never filmed, neither the cloud nor the mesh may contain the correct underside. Their agreement there says little. This is why camera coverage and silhouettes are included, and why a reference scan is needed for actual accuracy.

SECTION 06

From a weak region to a new capture angle

The capture compass turns an uncertainty map into a concrete suggestion for the next video or photograph.

For each weak region, TwinRecon considers candidate azimuth and elevation directions. It prefers a direction that appears able to see the region and is different from the camera angles already captured. This is an interpretable angular-coverage heuristic, inspired by next-best-view research. [5]



What recapture comparison currently means

A later scan can be compared against the earlier scan using aggregate counts and atlas summaries. The two reconstructions are evaluated separately because independent SfM runs can have different scales, orientations, and origins. TwinRecon does not yet perform exact point-by-point alignment between a before and after scan.

Azimuth means the horizontal direction around the object; **elevation** means how high or low the camera is. A useful new view sees a weak region clearly while still overlapping enough known surface for reliable camera alignment.

What a better recapture should look like

- Bring the camera to the suggested side and elevation while keeping good overlap with existing views.
- Keep the object rigid and sharp; use even light and a background that contrasts with the object.
- Capture the uncertain surface clearly, then rerun and compare its diagnostics with the original scan.

Current limit

The compass suggests a promising view; it does not guarantee improvement or implement the full mathematical optimization of POp-GS. A future experiment should compare its choices against random extra views.

SECTION 07

Problems encountered - and what changed

These are the practical issues that motivated the current design. Separate display problems from missing or incorrect 3D data.

'20% accuracy' sounded disastrous

The number was approximately 20.8% cloud-to-mesh disagreement under one distance threshold, not ground-truth accuracy. The interface and reports now use explicit diagnostic names. Real accuracy remains unmeasured without a reference scan.

The cloud looked blobby

Large browser dots hid fine shape. Point rendering was made smaller and clearer. Gaussian parameters now come from local oriented ellipsoids. This improves presentation and footprint shape but cannot invent absent surface detail.

The model appeared upside down

SfM coordinates have no natural 'up'. The viewer derives a display orientation from registered camera directions and applies it to the point cloud and consistency view. It does not rewrite the underlying geometry.

The splat looked too smooth

Oversized, round splats and display blending softened edges. Local ellipsoid scales, the browser Edge Detail control, and optional validation-gated refinement address this. Source blur and missing geometry remain limiting factors.

Display error versus reconstruction error

The upside-down orientation and large point dots were mainly viewer issues. Missing surfaces, floaters, and wrong camera poses are reconstruction issues. Fixing the display does not change the underlying 3D evidence.

SECTION 08

More problems encountered - and what changed

Difficult materials, foreground separation, appearance optimization, and hardware limits all affect the result.

Dark soles were cut off

Masks expand slightly around thin or dark boundaries. There is no final percentile box crop. RANSAC removes a plane only when its footprint is much larger than the rest of the object, protecting broad flat soles.

Background points leaked into the model

Object masks constrain early feature extraction and later 3D foreground voting. Both are needed because one bad mask or stereo estimate can leave support-surface points.

Colors could bleach during optimization

The optional gsplat pass starts from measured color, compares foreground radiance rather than black-background pixels, limits updates, and accepts changes only if held-out images improve.

The 4 GB GPU could run out of memory

Frame size and count are capped. Geometry and atlas work run mostly on CPU. The optional CUDA step limits Gaussian count, render size, and iterations; it skips safely if CUDA or enough free memory is unavailable.

Viewing required SuperSplat

The local website now renders exported Gaussian ellipses in WebGL2 and can display the point cloud and diagnostics. Its current renderer shows DC color only, so learned SH would require a compatible viewer for visual inspection.

What remains imperfect

GrabCut can include shadows; a smooth mesh can conceal missing geometry; the browser does not yet display higher-order SH; and the short CUDA refinement has not been verified on the existing scan.

SECTION 09

What the current numbers do - and do not - prove

Use the existing 'real' scan as a worked example, not as a universal quality guarantee.

50	482,919	350,000
registered camera frames	filtered cloud points	exported splats

20.8%

Older point-to-Poisson-mesh disagreement fraction. Points farther than a chosen scene-unit threshold were counted.

18.78%

Atlas-sampled points belonging to regions with confidence below the chosen 0.8 threshold.

Neither number is 'accuracy'

One number measures disagreement between related 3D outputs; the other measures how much of the scored cloud received a low internal confidence score. Both depend on thresholds and design choices. Neither compares with real-world ground truth.

How to measure real accuracy

- Get an independent, trusted reference 3D scan of the same object.
- Align the reconstruction to the reference, including orientation and scale.
- Measure surface distance and coverage, such as a Chamfer-style distance and an F-score at a stated distance tolerance.
- For appearance, compare renders of held-out camera views with real frames; also record peak VRAM and run time.

Interpretation: geometry metrics ask 'is the shape in the right place?' Image metrics ask 'does the model render like the object?' A good result should be evaluated on both.

Provenance: the worked-example counts and percentages come from the current scan's reconstruction report and confidence-atlas report. They describe one run and should not be generalized to all objects.

SECTION 10

Milestones, claims, and next experiments

The current project is a working pipeline with an explainable research prototype. Stronger claims need controlled evaluation.

Milestone 1 - working system

Achieved for the demonstrated scan: video to registered cameras, dense object cloud, Gaussian PLY, mesh, reports, and local web inspection. Reconstruction quality remains capture-dependent.

Milestone 2 - consistency-based reliability

Prototype implemented: regional camera, mask, image, and mesh evidence; reason codes; floater candidates; frame audit. Whether confidence predicts real geometric error has not yet been validated.

Milestone 3 - targeted recapture

Partly implemented: suggested view directions and aggregate before/after summaries. Automatic recapture, common-frame alignment, and verified gains over random viewpoints remain future work.

A defensible research experiment

- Scan several objects with an independent reference. Include dark, shiny, textured, and flat surfaces.
- Test whether atlas confidence predicts reference-scan error region by region; plot error against score.
- Remove each evidence signal in turn to learn which parts of the atlas actually help.
- Capture extra views selected by the compass and by a random baseline; compare geometry, appearance, and resource use.

The safest project claim

TwinRecon reconstructs an object from video and supplies an explainable estimate of weak regions. It does not yet establish a measured percentage of geometric accuracy or prove optimal recapture.

SECTION 11

How to present it to your teacher

Use this script first, then answer questions with the short explanations below.

Two-minute presentation script

TwinRecon starts with a video of one object. It selects frames and masks the object so the background does not dominate reconstruction. COLMAP matches features to estimate the camera position for each frame and builds sparse 3D points. Multi-view stereo estimates depth more densely. I keep cloud points supported by object masks across several views, then produce a Gaussian splat for appearance and a Poisson mesh for a connected surface. The research part is a confidence atlas. It combines camera support, silhouette agreement, image agreement, and cloud-to-mesh distance for each local region. It flags weak regions and suggests a new viewing angle. Its percentages are internal diagnostics, not ground-truth accuracy. The next scientific step is to compare its predictions with a trusted reference scan and test whether suggested views improve the model more than random views.

Why use a video? Many overlapping views make it possible to infer depth and camera movement from ordinary 2D images.

Why both a splat and a mesh? The splat is optimized for appearance; the mesh is a connected geometric surface and a useful consistency check.

Does 18.78% weak mean 81.22% accurate? No. It is a thresholded fraction of internally scored regions, with no independent ground truth.

What is the biggest limitation? A missing surface may be absent from every output, so internal comparisons can overlook it. Difficult capture and masks also limit quality.

Theory references

[1] COLMAP tutorial - colmap.github.io/tutorial.html
[2] Kerbl et al., 3D Gaussian Splatting (2023) - arxiv.org/abs/2308.04079
[3] Kazhdan et al., Poisson Surface Reconstruction (2006) - diglib.eg.org/items/245d6ba1-0209-4365-a4a4-16ce46421620
[4] gsplat 1.5.3 rasterization documentation - docs.gsplat.studio/versions/1.5.3/apis/rasterization.html
[5] POp-GS next-best-view paper (CVPR 2025) - arxiv.org/abs/2503.07819